

Reviewer Pack — Algebraic Topology of State Spaces Bounds Monotone Circuit S...

Entry 824d4bf2f928 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Algebraic Topology of State Spaces Bounds Monotone Circuit Size

Entry ID: 824d4bf2f928 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-29 03:39:47 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Topology (State Spaces) **Field B** (complexity object): Boolean Function Complexity (Monotone Circuit Size)

Statement:

[‘For every monotone Boolean function $f: \{0,1\}^n \rightarrow \{0,1\}$, let π_f be the fundamental group of the state space associated with f . Then the monotone circuit size for f is upper bounded by $2^{(\pi_f(n))}$.’, ‘For all $n \geq 1$ and all monotone Boolean functions f , there exists a polynomial-time computable invariant π_f such that the monotone circuit size for f satisfies $C_f \leq 2^{(\pi_f(n))}$ where C_f is the smallest monotone circuit computing f .’]

Rationale (proposer’s reasoning):

[‘Algebraic topology has been used to study complexity classes in some contexts, such as topological quantum computation. The state space of a Boolean function provides an abstract topological structure that might capture essential properties relevant to circuit complexity.’, ‘The fundamental group as an algebraic invariant is expected to reveal non-trivial structural information about the state spaces, potentially leading to new insights into circuit lower bounds.’]

Taxonomy category: TSEITIN_RES (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 68ee72ab9845c082

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a given monotone Boolean function, if the computed fundamental group π_f is such that $C_f \leq 2^{(\pi_f(n))}$, where C_f is the monotone circuit size and n is the number of variables, then the conjecture is supported.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.90	SAFE	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 2 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Algebraic Topology" AND "State Spaces" AND Boolean Function Complexity - "Monotone Circuit Size" AND "fundamental group" AND state space - "polynomial-time computable invariant" AND circuit complexity bound

Top relevant hits considered: - [s2:10.1007/978-3-642-13182-0_37] An Encoding Invariant Version of Polynomial Time Computable Distributions - [s2:2209.09400] Polynomial-Time Reachability for LTI Systems With Two-Level Lattice Neural Network Controllers

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.5s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def is_monotone(f):
        for i in range(1 << len(f)):
            for j in range(i + 1, 1 << len(f)):
                if f[i] > f[j]:
                    return False
            return True

    def fundamental_group(n):
        # Placeholder function to compute the fundamental group of a state space.
        # This is a stub and should be replaced with an actual algorithm.
        return random.randint(1, 5)

    def monotone_circuit_size(f):
        # Placeholder function to compute the monotone circuit size.
        # This is a stub and should be replaced with an actual algorithm.
        return len(f) * 2

    n = random.choice([5, 10, 15, 20, 30, 40])
    f = [random.randint(0, 1) for _ in range(1 << n)]
```

```

if not is_monotone(f):
    return {
        "metric_name": "monotone_circuit_size",
        "metric_value": None,
        "instances_tested": 1,
        "conjecture_holds": False,
        "counterexample": "not_monotone"
    }

pi_f = fundamental_group(n)
C_f = monotone_circuit_size(f)

if C_f <= 2 ** pi_f:
    conjecture_holds = True
    counterexample = ""
else:
    conjecture_holds = False
    counterexample = f"C_f={C_f} > 2^π_f({pi_f})={2**pi_f}"

return {
    "metric_name": "monotone_circuit_size",
    "metric_value": C_f,
    "instances_tested": 1,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    if len(sys.argv) > 1:
        seeds = [int(s) for s in sys.argv[1:]]
    else:
        seeds = [2**i + 1 for i in range(5, 8)] # First 30 prime numbers

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    if all(r["conjecture_holds"] for r in results):
        mean_value = sum(r["metric_value"] for r in results) / len(results)
        std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))
        support_fraction = 1.0
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        counterexample = next(result["counterexample"] for result in results if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TRIAL: {'metric_name': 'monotone_circuit_size', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'not_monotone'}

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80531895.py", line 76, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80531895.py", line 41, in run_trial
    if not is_monotone(f):
           ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80531895.py", line 24, in is_monotone
    if f[i] > f[j]:
       ~^^
IndexError: list index out of range
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from verifying whether the conjecture holds or not. | next: Investigate and fix the crash in the test code to allow for a proper verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 12

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15026
2	propose	ollama_remote	glm4:latest	0	0	9095
3	propose	ollama_remote	glm4:latest	0	0	11016
4	propose	ollama_remote	glm4:latest	0	0	10533
5	preregistration	ollama_remote	glm4:latest	0	0	5550
6	novelty	ollama_remote	glm4:latest	0	0	4818
7	novelty	ollama_remote	glm4:latest	0	0	11293
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17004
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7377
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8251
11	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8377
12	judge	ollama_remote	glm4:latest	0	0	13003

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 121344 ms total latency. Provider mix: {'ollama_remote': 12}

(full prompt+response transcripts available in [research/audit/824d4bf2f928.jsonl](#))

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/824d4bf2f928.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/824d4bf2f928.tar.gz` (if generated)